A Trace and Companion-Matrix Solution to the July 2010 IBM *Ponder This* Challenge

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Abstract

We solve the July 2010 IBM *Ponder This* challenge asking for a natural number n such that

$$10^9 \mid \mathrm{round}\big((1+2\cos 20^\circ)^n\big).$$

The central observation is that

$$\alpha = 1 + 2\cos 20^\circ$$

is a cubic Pisot unit: it is an algebraic integer greater than 1, while its other two conjugates have absolute value less than 1. Their powers complete α^n to the integer trace $\mathrm{Tr}(\alpha^n)$, with an error that decays exponentially. A companion matrix then turns the trace recurrence into finite modular arithmetic. An order calculation modulo 2 and 5, followed by lifting to 2^9 and 5^9, gives an explicit solution.

Challenge (IBM *Ponder This*, July 2010). The July 2010 IBM *Ponder This* challenge asks for a natural number n such that

$$10^9 \mid \mathrm{round}\big((1+2\cos 20^\circ)^n\big).$$

Here $\mathrm{round}(x)$ denotes the nearest integer to x. In the argument below, the number being rounded will always be strictly within $1/2$ of a unique integer, so no convention about half-integers is needed.

1 Introduction and strategy

The operation to simplify is rounding a very large real number. Seek an exact integer sequence that differs from α^n only by terms tending to zero. Adding the powers of the other conjugates supplies such a sequence; a recurrence will then let us calculate it modulo 10^9.

Set

$$\alpha = 1 + 2\cos 20^\circ = 1 + 2\cos\frac{\pi}{9}.$$

The key point is that α is a cubic algebraic integer whose other two conjugates lie strictly inside the unit circle. Thus, if β and γ are the other conjugates, then

$$\alpha^n + \beta^n + \gamma^n$$

is an integer, while $\beta^n + \gamma^n$ rapidly tends to 0. Therefore, for all sufficiently large n,

$$\mathrm{round}(\alpha^n) = \alpha^n + \beta^n + \gamma^n.$$

The rounding in the problem is therefore naturally explained by the trace map.

The remaining task is to force this trace to be 0 modulo 10^9. The trace sequence satisfies a linear recurrence coming from the minimal polynomial

$$X^3 - 3X^2 + 1.$$

Equivalently, it is encoded by the companion matrix

$$A = \begin{pmatrix} 3 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Since $\det A = -1$, this matrix is invertible modulo every positive integer. The congruence problem becomes a finite group problem: find a power of A that is congruent to the identity modulo 10^9, then use the fact that the trace at index -1 is 0.

We will prove that

$$n = 21{,}699{,}999{,}999$$

works. This value is not claimed to be minimal; the point is to give a clean Galois-theoretic and group-theoretic certificate.

2 The cubic Pisot unit

Let

$$\alpha = 1 + 2\cos\frac{\pi}{9}.$$

We first determine its minimal polynomial and its conjugates.

Lemma 1. *The number α has minimal polynomial*

$$f(X) = X^3 - 3X^2 + 1.$$

Its three conjugates are

$$\alpha = 1 + 2\cos\frac{\pi}{9}, \qquad \beta = 1 + 2\cos\frac{5\pi}{9}, \qquad \gamma = 1 + 2\cos\frac{7\pi}{9}.$$

Moreover,

$$|\beta| < \frac{2}{3}, \qquad |\gamma| < \frac{2}{3}.$$

Proof. We use the identity

$$2\cos(3\theta) = (2\cos\theta)^3 - 3(2\cos\theta).$$

Taking $\theta = \pi/9$, we have $3\theta = \pi/3$, hence

$$(2\cos(\pi/9))^3 - 3(2\cos(\pi/9)) = 2\cos(\pi/3) = 1.$$

Since $\alpha - 1 = 2\cos(\pi/9)$, this gives

$$(\alpha - 1)^3 - 3(\alpha - 1) = 1.$$

Expanding,

$$\alpha^3 - 3\alpha^2 + 1 = 0.$$

Thus α is a root of

$$f(X) = X^3 - 3X^2 + 1.$$

The rational root test shows that f has no rational root, since neither 1 nor -1 is a root. Therefore f is irreducible over $\mathbb{Q}$, and so f is the minimal polynomial of α.

The same trigonometric identity also shows that

$$1 + 2\cos\frac{5\pi}{9} \quad \text{and} \quad 1 + 2\cos\frac{7\pi}{9}$$

are roots of f, because

$$3 \cdot \frac{5\pi}{9} = \frac{5\pi}{3}, \qquad 3 \cdot \frac{7\pi}{9} = \frac{7\pi}{3},$$

and both $\cos(5\pi/3)$ and $\cos(7\pi/3)$ equal $1/2$. These three roots are distinct, so they are precisely the three conjugates of α.

It remains to bound the two smaller conjugates. We compute

$$f\left(-\frac{2}{3}\right) = -\frac{17}{27} < 0, \qquad f(0) = 1 > 0,$$

so one root lies in $(-2/3, 0)$. Also

$$f(0) = 1 > 0, \qquad f\left(\frac{2}{3}\right) = -\frac{1}{27} < 0,$$

so one root lies in $(0, 2/3)$. Finally,

$$f(2) = -3 < 0, \qquad f(3) = 1 > 0,$$

so the remaining root lies in $(2, 3)$. Since

$$\alpha = 1 + 2\cos\frac{\pi}{9} \in (2, 3),$$

the other two conjugates must lie in $(-2/3, 0)$ and $(0, 2/3)$. Hence

$$|\beta| < \frac{2}{3}, \qquad |\gamma| < \frac{2}{3}.$$

□

From the cyclotomic viewpoint, if

$$\zeta = e^{\pi i/9},$$

then

$$\alpha = 1 + \zeta + \zeta^{-1}.$$

Thus α lies in the real cubic subfield of the 18-th cyclotomic field. The three conjugates above correspond to the three embeddings of this real cubic field. This is the Galois-theoretic source of the trace that appears next.

3 The trace sequence and the rounding step

Let

$$K = \mathbb{Q}(\alpha).$$

For $n \in \mathbb{Z}$, define

$$S_n = \mathrm{Tr}_{K/\mathbb{Q}}(\alpha^n) = \alpha^n + \beta^n + \gamma^n.$$

Lemma 2. *For every $n \in \mathbb{Z}$, one has*

$$S_n \in \mathbb{Z}.$$

Moreover, for every $n \geq 4$,

$$\mathrm{round}(\alpha^n) = S_n.$$

Proof. Since f is monic, α is an algebraic integer. Also,

$$\alpha^3 - 3\alpha^2 + 1 = 0$$

implies

$$\alpha^{-1} = 3\alpha - \alpha^2.$$

Thus α is a unit. Hence α^n is an algebraic integer for every $n \in \mathbb{Z}$, and therefore its trace

$$S_n = \mathrm{Tr}_{K/\mathbb{Q}}(\alpha^n)$$

is an integer.

For $n \geq 4$, we have

$$|\alpha^n - S_n| = |\beta^n + \gamma^n| \leq |\beta|^n + |\gamma|^n < 2\left(\frac{2}{3}\right)^n \leq 2\left(\frac{2}{3}\right)^4 = \frac{32}{81} < \frac{1}{2}.$$

Since S_n is an integer, this proves

$$\mathrm{round}(\alpha^n) = S_n$$

for all $n \geq 4$. □

Thus the original problem has been reduced to the following congruence problem:

$$\text{Find } n \geq 4 \text{ such that } S_n \equiv 0 \pmod{10^9}.$$

4 The companion matrix

The trace sequence satisfies a recurrence coming from the polynomial

$$f(X) = X^3 - 3X^2 + 1.$$

Indeed, every root $r \in \{\alpha, \beta, \gamma\}$ satisfies

$$r^3 = 3r^2 - 1.$$

Multiplying by r^{n-3} and summing over the three conjugates gives

$$S_n = 3S_{n-1} - S_{n-3}.$$

It is convenient to encode this recurrence using the companion matrix

$$A = \begin{pmatrix} 3 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Then

$$\det(TI - A) = T^3 - 3T^2 + 1 = f(T).$$

Therefore the eigenvalues of A are α, β, γ, and for every $n \geq 0$,

$$\operatorname{tr}(A^n) = \alpha^n + \beta^n + \gamma^n = S_n.$$

We will also need the trace at index -1. Since

$$\det A = -1,$$

the matrix A is invertible over $\mathbb{Z}$, and a direct computation gives

$$A^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 3 & 0 \end{pmatrix}.$$

Hence

$$\operatorname{tr}(A^{-1}) = 0.$$

This agrees with

$$S_{-1} = \alpha^{-1} + \beta^{-1} + \gamma^{-1} = 0,$$

which also follows from the elementary symmetric relation

$$\alpha\beta + \alpha\gamma + \beta\gamma = 0.$$

So our goal is to find a large positive L such that

$$A^L \equiv I \pmod{10^9}.$$

Then

$$A^{L-1} \equiv A^{-1} \pmod{10^9},$$

and therefore

$$S_{L-1} = \operatorname{tr}(A^{L-1}) \equiv \operatorname{tr}(A^{-1}) = 0 \pmod{10^9}.$$

5 Orders modulo 2 and 5

We now compute suitable orders of A modulo 2 and 5.

Proposition 1. *Modulo 2, the matrix A has order 7. Modulo 5, the matrix A has order 62. In particular,*

$$A^7 \equiv I \pmod{2}, \qquad A^{62} \equiv I \pmod{5}.$$

Proof. Let

$$f(T) = T^3 - 3T^2 + 1.$$

For a prime p, let $\overline{f}_p$ denote the reduction of f modulo p, and let $\overline{A}_p$ denote the reduction of A modulo p.

First consider $p = 2$. We have

$$\overline{f}_2(T) = T^3 + T^2 + 1.$$

This polynomial has no root in $\mathbb{F}_2$, since

$$\overline{f}_2(0) = 1, \qquad \overline{f}_2(1) = 1.$$

As a cubic with no root, it is irreducible over $\mathbb{F}_2$.

Since $\overline{f}_2$ is the characteristic polynomial of $\overline{A}_2$, and is irreducible of degree 3, it is also the minimal polynomial of $\overline{A}_2$. Thus

$$\mathbb{F}_2[\overline{A}_2] \cong \mathbb{F}_2[T]/(\overline{f}_2) \cong \mathbb{F}_8.$$

Under this isomorphism, $\overline{A}_2$ corresponds to the class $\lambda = T \bmod \overline{f}_2$. Since $\lambda \neq 0, 1$, and $\mathbb{F}_8^\times$ has order 7, we get

$$\lambda^7 = 1$$

and λ has order 7. Therefore

$$\overline{A}_2^7 = I.$$

Now consider $p = 5$. We have

$$\overline{f}_5(T) = T^3 + 2T^2 + 1.$$

Checking the five elements of $\mathbb{F}_5$,

$$\overline{f}_5(0), \overline{f}_5(1), \overline{f}_5(2), \overline{f}_5(3), \overline{f}_5(4) = 1, 4, 2, 1, 2,$$

so $\overline{f}_5$ has no root in $\mathbb{F}_5$. Hence it is irreducible over $\mathbb{F}_5$.

Again,

$$\mathbb{F}_5[\overline{A}_5] \cong \mathbb{F}_5[T]/(\overline{f}_5) \cong \mathbb{F}_{125}.$$

Let $\lambda = T \bmod \overline{f}_5$. The norm of λ from $\mathbb{F}_{125}$ to $\mathbb{F}_5$ is

$$N_{\mathbb{F}_{125}/\mathbb{F}_5}(\lambda) = \lambda^{1+5+25} = \lambda^{31}.$$

On the other hand, since λ is a root of the monic cubic $\overline{f}_5(T) = T^3 + 2T^2 + 1$, the product of its three conjugates is -1. Therefore

$$\lambda^{31} = -1.$$

It follows that

$$\lambda^{62} = 1.$$

Moreover, $\lambda^{31} \neq 1$, and $\lambda \neq -1$, since -1 is not a root of $\overline{f}_5$. The only divisors of 62 are $1, 2, 31, 62$, so λ has order 62. Therefore

$$\overline{A}_5^{62} = I.$$

□

6 Lifting to 2^9 and 5^9

We use a standard lifting lemma.

Lemma 3. *Let p be a prime, and let B be an integer matrix satisfying*

$$B \equiv I \pmod{p}.$$

Then, for every $s \geq 0$,

$$B^{p^s} \equiv I \pmod{p^{s+1}}.$$

Proof. We argue by induction on s. The case $s = 0$ is the hypothesis.

Assume

$$B^{p^s} = I + p^{s+1}C$$

for some integer matrix C. Then

$$B^{p^{s+1}} = \left(B^{p^s}\right)^p = \left(I + p^{s+1}C\right)^p.$$

Expanding,

$$\left(I + p^{s+1}C\right)^p = I + p^{s+2}C + \sum_{j=2}^{p} \binom{p}{j} p^{j(s+1)} C^j.$$

Every term after I is divisible by p^{s+2}. Hence

$$B^{p^{s+1}} \equiv I \pmod{p^{s+2}}.$$

This completes the induction. □

Applying the lemma to $B = A^7$ with $p = 2$, we get

$$A^{7 \cdot 2^8} \equiv I \pmod{2^9}.$$

Applying it to $B = A^{62}$ with $p = 5$, we get

$$A^{62 \cdot 5^8} \equiv I \pmod{5^9}.$$

Now set

$$L = \operatorname{lcm}(7 \cdot 2^8,\ 62 \cdot 5^8).$$

Since $62 = 2 \cdot 31$, this least common multiple is

$$L = 2^8 \cdot 5^8 \cdot 7 \cdot 31 = 217 \cdot 10^8 = 21{,}700{,}000{,}000.$$

Thus

$$A^L \equiv I \pmod{2^9}, \qquad A^L \equiv I \pmod{5^9}.$$

Because 2^9 and 5^9 are coprime, the Chinese remainder theorem gives

$$A^L \equiv I \pmod{10^9}.$$

We can now finish the proof.

Theorem 1. *The integer*

$$\boxed{n = 21{,}699{,}999{,}999}$$

satisfies

$$10^9 \mid \operatorname{round}((1 + 2\cos 20^\circ)^n)\,.$$

Proof. Let

$$L = 21{,}700{,}000{,}000.$$

We have just shown that

$$A^L \equiv I \pmod{10^9}.$$

Therefore

$$A^{L-1} \equiv A^{-1} \pmod{10^9}.$$

Taking traces,

$$\operatorname{tr}(A^{L-1}) \equiv \operatorname{tr}(A^{-1}) \pmod{10^9}.$$

But

$$\operatorname{tr}(A^{L-1}) = S_{L-1}$$

and

$$\operatorname{tr}(A^{-1}) = 0.$$

Hence

$$S_{L-1} \equiv 0 \pmod{10^9}.$$

Since $L - 1 \geq 4$, the rounding lemma gives

$$\operatorname{round}(\alpha^{L-1}) = S_{L-1}.$$

Thus

$$10^9 \mid \operatorname{round}(\alpha^{L-1}).$$

Substituting

$$\alpha = 1 + 2\cos 20^\circ$$

and

$$L - 1 = 21{,}699{,}999{,}999$$

gives

$$10^9 \mid \operatorname{round}\Big((1 + 2\cos 20^\circ)^{21{,}699{,}999{,}999}\Big)\,.$$

Therefore

$$\boxed{n = 21{,}699{,}999{,}999}$$

is a valid answer. □

7 A coarser but very uniform variant

The exact order computation modulo 2 and 5 gives a reasonably small exponent. If one wants an even more uniform group-theoretic certificate, one can avoid computing the exact orders of A modulo 2 and 5.

Since

$$\det A = -1,$$

the reduction of A modulo any prime p lies in

$$\mathrm{GL}_3(\mathbb{F}_p).$$

Therefore the order of A modulo p divides

$$|\,\mathrm{GL}_3(\mathbb{F}_p)| = (p^3-1)(p^3-p)(p^3-p^2).$$

For $p = 2$,

$$|\,\mathrm{GL}_3(\mathbb{F}_2)| = (8-1)(8-2)(8-4) = 168.$$

For $p = 5$,

$$|\,\mathrm{GL}_3(\mathbb{F}_5)| = (125-1)(125-5)(125-25) = 1{,}488{,}000.$$

Thus

$$A^{168} \equiv I \pmod 2, \qquad A^{1,488,000} \equiv I \pmod 5.$$

The same lifting lemma gives

$$A^{168\cdot 2^8} \equiv I \pmod{2^9}, \qquad A^{1,488,000\cdot 5^8} \equiv I \pmod{5^9}.$$

Hence

$$M = 2^8 5^8 \cdot 168 \cdot 1{,}488{,}000 = 10^8 \cdot 168 \cdot 1{,}488{,}000 = 24{,}998{,}400{,}000{,}000{,}000$$

satisfies

$$A^M \equiv I \pmod{10^9}.$$

Exactly as before,

$$A^{M-1} \equiv A^{-1} \pmod{10^9},$$

so

$$S_{M-1} \equiv 0 \pmod{10^9}.$$

Since $M - 1 \geq 4$, we again have

$$\mathrm{round}(\alpha^{M-1}) = S_{M-1}.$$

Therefore the coarser group-order argument gives the valid exponent

$$\boxed{n = 24{,}998{,}399{,}999{,}999{,}999}.$$

This version uses only the general finite group bound

$$\mathrm{ord}(A \bmod p) \mid |\,\mathrm{GL}_3(\mathbb{F}_p)|.$$

The sharper proof above replaces those large group orders by the actual orders 7 and 62 of this particular companion matrix modulo 2 and 5.

8 Summary of the mechanism

The proof rests on three linked ideas.

First, the number

$$\alpha = 1 + 2\cos 20^\circ$$

is not just a trigonometric curiosity. It is a cubic algebraic integer with minimal polynomial

$$X^3 - 3X^2 + 1.$$

Its other two conjugates have absolute value less than 1, so powers of α are very close to the integer traces

$$\mathrm{Tr}_{K/\mathbb{Q}}(\alpha^n).$$

This explains the appearance of the rounding function.

Second, the trace sequence

$$S_n = \mathrm{Tr}_{K/\mathbb{Q}}(\alpha^n)$$

satisfies the same recurrence dictated by the minimal polynomial:

$$S_n = 3S_{n-1} - S_{n-3}.$$

The companion matrix packages this recurrence into the identity

$$S_n = \mathrm{tr}(A^n).$$

Third, because the companion matrix is invertible modulo 10^9, its powers are periodic modulo 10^9. Choosing L so that

$$A^L \equiv I \pmod{10^9}$$

forces

$$A^{L-1} \equiv A^{-1} \pmod{10^9}.$$

The trace of A^{-1} is 0, so

$$S_{L-1} \equiv 0 \pmod{10^9}.$$

Since S_{L-1} is exactly the rounded value of α^{L-1}, the desired divisibility follows.